

AI, Productivity and Prices — Model Note

Margin of Error · Methods appendix · 2026

Abstract

This note documents the data sources, transformations and modelling choices behind The Observatory, an interactive tool that combines observed macroeconomic data with a small, stylised simulation model to illustrate how AI adoption might affect productivity, inflation, unemployment and the distribution of wage gains across worker groups.

The Observatory addresses a specific question: through which economic channels can AI adoption affect productivity, inflation, unemployment and the distribution of gains between higher-skill and lower-skill workers, and what do currently available data series suggest about whether those channels are active?

It does not attempt to produce a point forecast. Instead it pursues two complementary objectives. First, it presents observed data from public sources — the World Bank and the Federal Reserve Economic Data (FRED) repository — that correspond to the nodes of the AI-to-macro transmission chain: price indices, productivity measures, investment proxies and wage or earnings series. These are presented descriptively, without imposing a structural interpretation. Second, it provides a stylised simulation model that generates time paths for inflation, unemployment, the policy interest rate and real wages under user-specified assumptions about adoption speed, the wage-vs-profit split of productivity gains, monetary policy stance and whether AI tends to complement or displace labour.

The model is deliberately simple and non-predictive; all numerical results are illustrative. It is intended for readers who want to understand or replicate the underlying mechanics.

Data sources and transformations

2.1 World Bank series

The baseline dataset draws on two World Bank Development Indicators, available for all five countries in the atlas (United States, United Kingdom, Germany, Japan, Singapore):

Series	Indicator code	Transform
Labour productivity	NY.GDP.PCAP.KD.ZG	Annual % growth → 5-yr rolling mean
CPI inflation	FP.CPI.TOTL.ZG	Annual %, as reported
ICT investment	GB.XPD.RSDV.GD.ZS	% of GDP, proxy for AI capital spend
Employment rate	SL.EMP.TOTL.SP.ZS	Inverted to approximate unemployment

2.2 FRED enrichment (US only, optional)

When a FRED API key is present, the pipeline appends higher-frequency US series: PCE deflator (PCEPI, monthly), nonfarm business labour productivity (PR85006092, quarterly), ICT private fixed investment (Y033RC1Q027SBEA, quarterly) and real median weekly earnings (LES1252881600Q, quarterly). All series are interpolated to annual frequency and aligned to the World Bank calendar year.

Simulation model

The Lab section of The Observatory exposes a four-parameter simulation. The parameters and their interpretation are:

Parameter	Symbol	Range	Interpretation
Adoption speed	α	0 – 1	Annual share of tasks exposed to AI
Productivity gain	ϕ	0 – 5 %	Output per worker uplift per period

Wage-profit split	λ	0 – 1	Share of gains flowing to wages vs. profits
Monetary stance	ρ	0 – 2	Taylor-rule aggressiveness on inflation

Illustrative scenario: Rapid Adoption, Labour Capture

Setting $\alpha = 0.8$, $\phi = 3\%$, $\lambda = 0.7$ (70% of gains to wages), $\rho = 1$ (moderate monetary response) produces the following stylised trajectory over a ten-year horizon:

Year	Productivity (cumul.)	CPI inflation	Unemployment	Real wage (index)
0	—	2.0 %	4.5 %	100
2	+3.1 %	2.4 %	4.2 %	103
5	+8.6 %	1.8 %	3.8 %	109
10	+18.4%	1.4 %	3.5 %	119

Note: all values are model outputs under the specified parameter choices, not forecasts. Productivity is measured relative to a no-AI counterfactual. The model has no external sector, no credit channel, and no structural breaks — features that would matter in a realistic calibration.

What this is not

The Observatory is not a forecasting model. It does not produce probability distributions over future outcomes. It does not incorporate financial market data, supply-chain dynamics, geopolitical shocks, or sector-level heterogeneity in AI adoption. The observed data panels and the simulation model are parallel rather than integrated: the model does not re-estimate itself from the data. These limitations are intentional — the goal is transparency and learnability, not realism. A reader who wants a calibrated macro model should consult the IMF World Economic Outlook or the academic literature on AI and growth (Acemoglu & Restrepo 2019; Brynjolfsson, Rock & Syverson 2021).